

AC power-flow surrogates on the ppNR v2 corpus

Four models, 31 grids: results so far

PIGNN-Attn-LS-PPC experimental summary

19 August 2026

1 Scope and status

Four PF surrogates, one model per grid, trained on the regenerated **ppNR v2** corpus. v2 is selected by the `_u0clean_sgenpert_pvqvstart_` configuration tag rather than the `_ppNR_` solver tag: several generations now share the latter, so selecting on it would silently mix corpora. All four share the parquet, the 1/3–1/3–1/3 split and seed 42.

Tag	Model	Finished	Usable
<code>pignn</code>	<code>GNSMsg_EdgeSelfAttn</code> , unrolled solver $K = 40$	26/31	22
<code>graphkit</code>	IBM <code>gridfm_graphkit</code> <code>GNS_heterogeneous</code>	30/31	30
<code>gridsfm</code>	<code>gridsfm.GridTransformerBackbone</code> (released)	26/31	26
<code>lumina</code>	<code>lumina_inference</code> HGT (released)	30/31	30

Table 1: All trained from scratch; no released weights are loaded. “Usable” excludes runs whose validation RMSE never moved. Aggregates below are over the **22 grids where all four produced a usable model**.

Placement. Grids were assigned to hardware by size: the 10 smallest to alex a40, case39–case_illinois200 to alex2 a100, case300–case1888rte and the nine largest to helma h100. Batch sizes follow from that placement by the rule in Section 2.

PIGNN configuration used for the results. The `pignn` rows were launched by `dispatch_pf_v2.sh`; this table gives the complete effective setting (driver flags plus parser defaults).

Setting	Value
Model / architecture	<code>GNSMsg_EdgeSelfAttn</code> ; $d = 4$, $d_{hi} = 24$
Attention	8 heads; 8 attention layers
Unrolled solver	$K = 40$; $\gamma = 0.9$
PINN / batching	PINN enabled; block-diagonal batches
Solver step	Armijo enabled; <code>geometric_safe</code> ; $\rho = 0.5$, $c_1 = 10^{-4}$, 5 backtracks, $\alpha_{min} = 0.0625$
Objective	$1.0 L_{MSE} + 1.0 L_{physics}$; physics form <code>logcosh</code>
Physics normalization	residual norm <code>none</code> ; norm $\epsilon = 10^{-6}$
Optimization	AdamW; learning rate 10^{-4} ; weight decay 10^{-3}
Schedule / accumulation	default scheduler; accumulation 1; no effective-batch target
Training budget	40 epochs (largest-grid reruns used reduced budgets noted in Section 4)
Data split	train/valid/test = 1/3/1/3/1/3; seed 42
Data scaling / loading	per-unit; target $S_{base} = 100$ MVA; shared grid and Y-bus cache
Numerics	complex64 dataset tensors; no extra v_n or bus-type features

Table 2: Effective PIGNN settings for the `pignn` results. Batch size is grid-dependent and listed below.

2 Batch sizes and the rule that set them

Batch size was never chosen per grid by hand. It follows one formula, applied by `dispatch_pf_v2.sh`:

$$B(\text{model}, \text{grid}) = \min\left(\text{CAP}, \left\lfloor \frac{\text{BUDGET} \times M/48}{N} \right\rfloor\right),$$

where N is the bus count, M the GPU memory in GiB, and BUDGET a per-model allowance in *bus-samples* calibrated so that $B \cdot N$ activations fit a 48 GiB card with headroom. The budgets differ because the measured activation cost per bus-sample differs by a factor of seven across the four models.

Model	activation [KiB / bus-sample]	BUDGET [bus-samples @ 48 GiB]	CAP
pigmn	~1080	27 000	32
graphkit	~479	60 000	64
gridsfm	~1040	30 000	32
lumina	~145	200 000	32

Table 3: Every budget lands at roughly 28 GiB of activations on a 48 GiB card, i.e. about 60% of it. M was 48 (alex a40), 40 (alex2 a100), or 94 (helma h100) depending on where the grid ran.

Grid	Buses	pigmn	graphkit	gridsfm	lumina
case4gs	4	32	64	32	32
case5	5	32	64	32	32
case6ww	6	32	64	32	32
case9	9	32	64	32	32
case14	14	32	64	32	32
case24_ieee_rts	24	32	64	32	32
GBreducednetwork	29	32	64	32	32
case30	30	32	64	32	32
case_ieee30	30	32	64	32	32
case33bw	33	32	64	32	32
case39	39	32	64	32	32
case57	57	32	64	32	32
case89pegase	89	32	64	32	32
SimBench	94	32	64	32	32
case118	118	32	64	32	32
case145	145	32	64	32	32
iceland	189	32	64	32	32
case_illinois200	200	32	64	32	32
case300	300	32	64	32	32
LVN_heo1	722	32	64	32	32
case1354pegase	1354	32	64	32	32
case1888rte	1888	28	62	31	32
GBnetwork	2224	23	52	26	32
case2848rte	2848	18	41	20	32
case2869pegase	2869	18	40	20	32
case3120sp	3120	16	37	18	32
ENTSO_E_RealGridTest	6051	8	19	9	32
case6470rte	6470	8	18	9	32
case6495rte	6495	8	18	9	32
case6515rte	6515	8	18	9	32
case9241pegase	9241	5	12	6	32

Table 4: Micro-batch actually used, in graphs per optimiser step. **Bold** marks a grid where memory bound the batch below the cap. case1888rte first ran on alex a100_80 ($M=80$, batches 23/52/31) and was rerun on helma; the helma figures are shown.

The cap binds up to case1354pegase; past it, memory does. Every model runs at its cap on the first 21 grids, so nothing about batch size distinguishes them there. From case1888rte on, B falls with $1/N$: by case9241pegase pigmn is at 5 graphs per step, gridsfm at 6 and graphkit at 12 — reductions of $6.4\times$, $5.3\times$ and $5.3\times$ against their small-grid batches.

This confounds the size comparison. Gradient noise scales as $1/\sqrt{B}$, so the largest grids were optimised under roughly 2.5 times more noise than the smallest, at the same learning rate and the same 40-epoch budget. Any degradation reported above for $N \geq 2000$ mixes the difficulty of the grid with a weaker optimiser setting, and the two are not separated by these runs.

`lumina` is the exception: its activation cost is low enough that its budget never binds, so it held $B = 32$ on all 31 grids. It is the only model here whose optimiser saw a constant batch across the corpus — and also the only one whose error is flat in N , which is consistent with it being limited by something other than batch size.

This is now adjustable. All four drivers take `--grad_accum_steps` and `--target_effective_batch`; the latter derives the accumulation count from `--BATCH` so the effective batch can be held fixed across grids without more memory. Gradient clipping moved to once per accumulation group. Neither flag was used for the runs reported here, all of which ran at accumulation 1.

3 Two failure modes, both identified

Ten LUMINA runs died on a missing file. Every LUMINA job on the alex tier failed at `make_model` with `FileNotFoundError` on `lumina_ckpt/lumina_config.json`. LUMINA reads its architecture from the released config even when training from scratch, and the file existed only under the `b313dc11` account while the dispatcher built the path from `$HOME_BASE`. Copying it to the `iwi5295h` account and resubmitting fixed all ten; they now train normally.

Twelve runs hit the 24-hour wall clock, and the fix for it failed. All twelve are among the five largest grids, where a single epoch costs 2900–6700 s, so 40 epochs needs 32–74 h:

Model	Grid	epochs reached	s/epoch	40 epochs needs
pigmn	ENTSO_E_RealGridTest	29/40	2920	32.4 h
pigmn	case9241pegase	17/40	5125	57.0 h
graphkit	case9241pegase	28/40	3087	34.3 h
gridsfm	case6470rte	24/40	3602	40.0 h
gridsfm	case9241pegase	13/40	6655	74.0 h
lumina	case9241pegase	35/40	2450	27.2 h

Table 5: Six of the twelve.

`helma`’s `preempt` partition allows 48 h on the same `h100s`, so all twelve were resubmitted there. **Every one was evicted within three hours** (elapsed 2:44 to 3:03, state `PREEMPTED`), and since the drivers have no resume, that work was lost outright — a worse outcome than the 24 h runs it was meant to replace. Choosing `preempt` on its time limit alone, without weighting how contended it is, was the wrong call.

They are now back on `h100` with the epoch budget cut to what fits 21 h at each run’s measured per-epoch cost: pigmn 14–25, graphkit 24, gridsfm 11–23, lumina 30, against 40 everywhere else. **Those twelve cells will therefore carry a smaller training budget than the rest of the report**, and are not equal-budget comparisons. They are still training and do not appear in the tables below.

4 Results

Grid	Buses	V RMSE [pu]			
		pignn	graphkit	gridsfm	lumina
case4gs	4	3.43e-05	3.22e-04	9.97e-03	1.80e-02
case5	5	1.62e-05	1.19e-04	1.55e-03	2.46e-02
case6ww	6	2.13e-05	4.54e-04	6.01e-03	1.59e-02
case9	9	1.12e-04	7.39e-04	4.42e-03	2.02e-02
case14	14	8.13e-05	9.98e-04	3.17e-03	2.26e-02
case24_ieee_rts	24	6.53e-05	1.24e-03	7.19e-03	2.49e-02
GBreducednetwork	29	4.01e-05	4.09e-01	5.25e-02	6.02e-02
case30	30	1.34e-04	1.13e-03	1.63e-03	2.20e-02
case_ieee30	30	2.22e-04	2.13e-03	3.04e-03	1.99e-02
case33bw	33	3.84e-03	1.45e-03	5.14e-03	1.78e-02
case39	39	4.57e-05	7.29e-03	8.92e-03	2.87e-02
case57	57	1.92e-02	4.59e-03	1.97e-02	3.60e-02
case89pegase	89	2.71e-03	1.08e-02	2.00e-02	2.96e-02
SimBench	94	1.52e-03	5.15e-03	1.14e-02	5.42e-02
case118	118	3.22e-05	1.31e-03	1.95e-02	2.52e-02
case145	145	9.29e-03	1.25e-01	4.80e-02	6.70e-02
iceland	189	1.05e-01	2.11e-02	4.63e-02	3.99e-02
case_illinois200	200	1.52e-04	8.09e-03	5.39e-03	2.38e-02
case300	300	2.63e-02	2.37e-02	2.47e-02	4.23e-02
LVN_heo1	722	NAN	2.02e-02	3.09e-02	3.09e-02
case1354pegase	1354	1.73e-01	2.09e-02	2.54e-02	2.99e-02
case1888rte	1888	NAN	2.52e-02	2.90e-02	3.29e-02
GBnetwork	2224	NAN	3.54e-02	3.19e-02	4.44e-02
case2848rte	2848	NAN	2.64e-02	1.67e-02	4.00e-02
case2869pegase	2869	1.77e-01	3.21e-02	1.74e-02	4.13e-02
case3120sp	3120	1.76e-01	2.86e-02	3.05e-02	3.37e-02
ENTSO_E_RealGridTest	6051	—	2.42e-02	2.70e-02	3.46e-02
case6470rte	6470	—	4.00e-02	3.29e-02	4.31e-02
case6495rte	6495	—	3.87e-02	3.54e-02	4.46e-02
case6515rte	6515	—	3.81e-02	3.91e-02	4.42e-02
case9241pegase	9241	—	5.32e-02	4.00e-02	6.02e-02

Table 6: Voltage magnitude RMSE [pu]; best per grid in bold. NAN marks a run that diverged (Section 4).

Grid	Buses	θ RMSE [deg]			
		pigmn	graphkit	gridsfm	lumina
case4gs	4	2.70e-03	2.47e-02	2.21e-01	5.58e-01
case5	5	5.37e-03	6.12e-02	2.89e-02	5.25e-01
case6ww	6	8.73e-03	6.54e-02	3.13e+00	8.05e-01
case9	9	5.26e-02	1.18e-01	6.82e+00	1.38e+00
case14	14	1.52e-02	1.30e-01	2.45e+00	1.17e+00
case24_ieee_rts	24	3.58e-02	2.64e-01	1.62e+00	3.03e+00
GBreducednetwork	29	1.18e-01	1.02e+01	3.27e+01	1.20e+01
case30	30	4.73e-02	9.80e-02	1.84e+00	8.98e-01
case_ieee30	30	1.71e-02	1.24e-01	3.86e+00	2.16e+00
case33bw	33	2.65e-01	1.02e-01	1.47e+00	2.94e-01
case39	39	1.84e-02	5.11e-01	1.88e+01	5.76e+00
case57	57	1.03e-01	2.27e-01	3.06e+00	3.69e+00
case89pegase	89	2.88e-01	6.82e-01	1.06e+00	6.58e+00
SimBench	94	9.96e-02	2.78e-01	4.86e+00	7.36e+00
case118	118	3.77e-02	4.58e-01	3.64e+01	3.66e+00
case145	145	4.54e+00	1.84e+01	1.01e+02	6.84e+01
iceland	189	4.20e+01	2.51e+00	2.55e+01	9.32e+00
case_illinois200	200	5.62e-02	5.51e-01	5.03e+01	2.53e+00
case300	300	2.69e+00	4.76e+00	2.77e+01	1.90e+01
LVN_heo1	722	NAN	1.35e+00	7.49e-01	1.83e+00
case1354pegase	1354	7.01e+01	4.73e+00	1.43e+01	1.05e+01
case1888rte	1888	NAN	7.18e+00	1.56e+02	1.44e+01
GBnetwork	2224	NAN	4.94e+00	2.14e+00	1.72e+01
case2848rte	2848	NAN	3.12e+00	6.97e+01	5.73e+00
case2869pegase	2869	7.56e+01	9.69e+00	7.04e+01	2.87e+01
case3120sp	3120	7.97e+01	8.03e+00	1.23e+01	1.64e+01
ENTSO_E_RealGridTest	6051	—	4.15e+00	1.03e+01	1.10e+01
case6470rte	6470	—	1.22e+01	3.67e+01	1.74e+01
case6495rte	6495	—	3.39e+00	3.53e+01	1.17e+01
case6515rte	6515	—	7.98e+00	1.04e+02	1.45e+01
case9241pegase	9241	—	3.71e+01	1.19e+02	6.58e+01

Table 7: Voltage angle RMSE [deg].

PIGNN leads on every family. Over the 22 grids all four completed:

median over 22 grids	pigmn	graphkit	gridsfm	lumina
$ V $ RMSE [pu]	1.87e-04	4.87e-03	1.07e-02	2.70e-02
θ RMSE [deg]	0.0779	0.368	5.84	3.68
$\text{mean}_s \ \Delta P_s\ _\infty$ [pu]	1.60e-02	2.34e-01	8.32e-01	3.01
$\text{mean}_s \ \Delta Q_s\ _\infty$ [pu]	1.18e-02	1.77e-01	3.92e-01	4.45e-01
$ \Delta P $ [pu]	1.75e-03	3.90e-02	1.00e-01	4.00e-01
$ \Delta Q $ [pu]	1.85e-03	3.03e-02	1.16e-01	1.35e-01
p95 $ \Delta P $ [pu]	4.40e-03	1.22e-01	4.39e-01	1.38
p95 $ \Delta Q $ [pu]	5.62e-03	9.34e-02	3.40e-01	3.67e-01
median $ \Delta P $ [pu]	8.49e-04	2.46e-02	3.33e-02	1.78e-01
median $ \Delta Q $ [pu]	9.62e-04	1.80e-02	7.21e-02	7.88e-02
frac $ \Delta P \leq 10^{-6}$	0.06%	0.00%	0.00%	0.00%
frac $ \Delta Q \leq 10^{-6}$	0.06%	0.00%	0.00%	0.00%

Table 8: Win counts follow the same order: PIGNN takes 14–17 of 22 on every family, graphkit 1–6, gridsfm 0–6, lumina **0 on all eight**. The median and tolerance-fraction rows are over the 20 grids where all four models have a rescored distribution, so they are not directly comparable with the 22-grid rows above.

The ordering is **pigmn** < **graphkit** < **gridsfm** < **lumina** on all eight metrics, with no crossing. PIGNN’s margin is largest on the residuals — a factor of 15 against graphkit on $\text{mean}_s \|\Delta P_s\|_\infty$ and 22 on $|\Delta P|$ — which is the expected shape for a model whose iteration is driven by that residual.

Grid	Buses	mean _s $\ \Delta P_s\ _\infty$ [pu]				mean _s $\ \Delta Q_s\ _\infty$ [pu]			
		pigmn	graphkit	gridsfm	lumina	pigmn	graphkit	gridsfm	lumina
case4gs	4	1.15e-03	1.37e-02	7.23e-02	3.24e-01	1.97e-03	2.00e-02	2.01e-01	1.86e-01
case5	5	3.52e-03	5.35e-02	3.50e-02	4.89e-01	3.73e-03	2.40e-02	1.26e-01	1.28e-01
case6ww	6	5.37e-04	1.65e-02	4.91e-01	1.11e-01	4.10e-04	1.08e-02	1.51e-01	1.29e-01
case9	9	2.49e-03	2.20e-02	4.44e-01	1.99e-01	1.61e-03	1.75e-02	1.07e-01	9.96e-02
case14	14	8.07e-04	2.36e-02	5.51e-01	5.28e-01	7.05e-04	1.55e-02	5.22e-02	7.29e-02
case24_ieee_rts	24	2.81e-03	1.51e-01	5.89e-01	1.97e+00	3.85e-03	5.95e-02	3.10e-01	3.79e-01
GBreducednetwork	29	3.56e-01	4.66e+03	8.06e+01	2.10e+02	2.08e-01	2.98e+00	1.45e+00	5.46e+02
case30	30	2.53e-03	1.86e-02	1.73e-01	1.01e-01	2.20e-03	1.60e-02	9.75e-02	7.55e-02
case_ieee30	30	1.79e-03	4.04e-02	6.17e-01	3.68e-01	1.22e-03	3.04e-02	6.36e-02	1.02e-01
case33bw	33	2.21e-02	3.88e-03	3.88e-02	1.25e-01	2.61e-02	8.09e-03	3.13e-02	9.96e-02
case39	39	1.79e-03	3.71e-01	1.02e+00	4.46e+00	1.53e-03	3.43e-01	4.51e-01	8.36e-01
case57	57	3.97e-02	5.92e-02	2.16e+00	1.09e+00	4.89e-02	3.08e-02	3.34e-01	4.14e-01
case89pegase	89	1.55e+00	9.84e-01	6.45e-01	1.95e+01	2.64e+00	1.06e+00	7.91e-01	2.63e+00
SimBench	94	3.05e-02	2.48e-01	1.61e+00	7.15e+00	1.92e-02	2.37e-01	1.05e+00	3.48e+00
case118	118	9.88e-03	2.20e-01	2.38e+00	2.27e+00	4.29e-03	1.18e-01	7.08e-01	4.75e-01
case145	145	1.01e+00	2.09e+02	6.46e+01	2.74e+03	2.78e-01	3.88e+01	2.48e+00	2.35e+03
iceland	189	7.23e+03	1.66e+00	3.39e-01	5.30e+00	1.19e+04	1.21e+00	1.22e+02	2.13e+00
case_illinois200	200	7.26e-03	2.56e-01	1.72e+00	3.75e+00	1.75e-03	2.67e-01	2.26e-01	3.12e-01
case300	300	1.68e+00	1.04e+01	1.22e+00	8.06e+01	4.58e+00	6.82e+00	1.23e+00	1.81e+01
LVN_heo1	722	NAN	5.64e+00	1.54e+00	5.23e+00	NAN	2.34e+00	3.81e+00	1.80e+00
case1354pegase	1354	1.17e+04	4.79e+01	1.81e+00	4.32e+02	1.37e+04	1.42e+01	1.49e+00	1.22e+01
case1888rte	1888	NAN	6.81e+01	2.59e+02	5.69e+02	NAN	4.64e+01	8.89e+01	1.18e+02
GBnetwork	2224	NAN	1.81e+02	1.10e+00	3.18e+04	NAN	5.58e+01	4.81e+00	3.95e+03
case2848rte	2848	NAN	1.31e+01	1.97e+01	2.10e+02	NAN	2.26e+01	2.20e+00	1.13e+02
case2869pegase	2869	1.16e+04	4.44e+02	2.53e+00	7.66e+03	1.65e+04	1.35e+02	1.22e+00	6.69e+03
case3120sp	3120	1.97e+04	1.94e+03	1.79e+03	7.97e+03	3.64e+04	8.85e+01	4.09e+01	4.18e+03
ENTSO_E_RealGridTest	6051	—	3.61e+01	9.48e+00	7.75e+01	—	1.91e+01	2.57e+01	4.34e+01
case6470rte	6470	—	1.66e+02	3.70e+01	1.08e+03	—	8.33e+01	4.00e+01	3.36e+02
case6495rte	6495	—	1.50e+03	3.58e+02	5.46e+03	—	2.72e+02	1.70e+02	3.17e+03
case6515rte	6515	—	1.12e+02	2.20e+02	9.71e+02	—	1.08e+02	8.51e+01	4.20e+02
case9241pegase	9241	—	2.05e+03	4.52e+00	8.56e+03	—	6.97e+02	1.99e+01	1.34e+04

Table 9: Mean per-scenario maximum power-balance residual, pu on a 100 MVA base. The reducer is $N_s^{-1} \sum_s \max_i |\Delta P_{s,i}|$ (and similarly for Q), not $\max_{s,i}$. P is not scored at the slack, Q not at slack or PV.

Residual reducer convention. The legacy log names dPinf/dQinf and the earlier notation $\Delta P^\infty/\Delta Q^\infty$ can suggest a global test-set maximum. The driver actually computes a maximum over buses for each scenario, sums those values, and divides by the number of scenarios. Therefore all values in the preceding table are *mean per-scenario maxima*. By contrast, $|\Delta P|$, RMSE and tolerance fractions pool all valid bus-scenario entries. Median/p95 are pooled quantiles (with at most two million uniformly spaced entries retained by the older GridFM-derived driver on very large arrays). A true global maximum $\max_{s,i} |\Delta P_{s,i}|$ is not reported in this document.

Grid	Buses	$ \overline{\Delta P} $ [pu]				$ \overline{\Delta Q} $ [pu]			
		pignn	graphkit	gridsfm	lumina	pignn	graphkit	gridsfm	lumina
case4gs	4	7.42e-04	7.92e-03	4.04e-02	2.11e-01	1.67e-03	1.63e-02	1.42e-01	1.30e-01
case5	5	2.37e-03	2.73e-02	1.90e-02	2.63e-01	3.73e-03	2.40e-02	1.27e-01	1.28e-01
case6ww	6	3.04e-04	8.25e-03	2.36e-01	5.39e-02	2.62e-04	6.22e-03	9.94e-02	7.76e-02
case9	9	1.04e-03	9.41e-03	1.09e-01	9.54e-02	9.22e-04	8.21e-03	4.61e-02	4.43e-02
case14	14	2.08e-04	7.64e-03	7.30e-02	1.03e-01	3.01e-04	6.38e-03	1.74e-02	3.04e-02
case24_ieee_rts	24	1.02e-03	4.08e-02	1.43e-01	4.94e-01	1.39e-03	2.19e-02	1.09e-01	1.40e-01
GBreducednetwork	29	5.99e-02	2.16e+02	4.38e+00	1.94e+01	9.08e-02	1.19e+00	5.66e-01	1.10e+02
case30	30	4.07e-04	5.06e-03	2.76e-02	2.53e-02	3.15e-04	4.45e-03	1.62e-02	1.31e-02
case_ieee30	30	2.10e-04	9.73e-03	4.43e-02	9.59e-02	2.64e-04	8.74e-03	1.34e-02	2.61e-02
case33bw	33	1.36e-03	9.12e-04	1.10e-02	1.40e-02	1.48e-03	1.32e-03	8.22e-03	9.69e-03
case39	39	3.31e-04	9.34e-02	2.56e-01	7.65e-01	5.32e-04	9.85e-02	1.23e-01	2.16e-01
case57	57	5.51e-03	1.24e-02	9.84e-02	2.14e-01	8.01e-03	8.05e-03	5.36e-02	6.78e-02
case89pegase	89	5.31e-02	1.90e-01	9.64e-02	1.72e+00	1.55e-01	1.76e-01	1.26e-01	5.50e-01
SimBench	94	2.13e-03	4.70e-02	4.02e-01	8.40e-01	2.03e-03	4.65e-02	1.57e-01	3.42e-01
case118	118	8.71e-04	3.71e-02	2.31e-01	3.06e-01	7.20e-04	2.26e-02	1.46e-01	9.07e-02
case145	145	5.24e-02	8.34e+00	3.67e+00	8.12e+01	2.73e-02	2.94e+00	2.74e-01	1.04e+02
iceland	189	2.31e+02	1.42e-01	2.89e-02	6.50e-01	3.66e+02	1.18e-01	9.22e-01	1.66e-01
case_illinois200	200	3.85e-04	3.73e-02	8.30e-02	2.60e-01	2.60e-04	3.66e-02	4.45e-02	5.71e-02
case300	300	3.71e-02	6.05e-01	9.80e-02	2.50e+00	8.65e-02	3.15e-01	1.06e-01	6.01e-01
LVN_heo1	722	NAN	1.04e-01	1.26e-02	3.34e-02	NAN	9.66e-02	1.14e-01	8.04e-02
case1354pegase	1354	3.72e+02	1.08e+00	1.02e-01	1.76e+00	5.28e+02	5.34e-01	1.16e-01	4.22e-01
case1888rte	1888	NAN	1.24e+00	6.77e-01	1.69e+00	NAN	7.06e-01	3.56e-01	4.49e-01
GBnetwork	2224	NAN	2.05e+00	8.28e-02	1.38e+02	NAN	7.71e-01	2.66e-01	3.18e+01
case2848rte	2848	NAN	4.44e-01	1.24e-01	7.78e-01	NAN	2.79e-01	1.44e-01	3.63e-01
case2869pegase	2869	3.89e+02	4.92e+00	1.23e-01	7.81e+01	6.71e+02	1.77e+00	1.16e-01	2.51e+01
case3120sp	3120	5.83e+02	1.78e+00	6.44e-01	8.07e+00	8.70e+02	6.13e-01	1.96e-01	2.14e+00
ENTSO_E_RealGridTest	6051	—	3.46e-01	5.33e-02	5.83e-01	—	1.88e-01	2.00e-01	2.38e-01
case6470rte	6470	—	1.04e+00	1.01e-01	1.80e+00	—	4.37e-01	2.36e-01	4.98e-01
case6495rte	6495	—	9.51e-01	4.65e-01	2.12e+00	—	3.41e-01	3.71e-01	9.53e-01
case6515rte	6515	—	7.27e-01	2.96e-01	1.52e+00	—	3.63e-01	3.06e-01	4.98e-01
case9241pegase	9241	—	1.12e+01	9.99e-02	2.58e+02	—	4.29e+00	3.75e-01	3.06e+02

Table 10: Mean residual, pu.

Grid	Buses	p95 $ \Delta P $ [pu]				p95 $ \Delta Q $ [pu]			
		pigmn	graphkit	gridsfm	lumina	pigmn	graphkit	gridsfm	lumina
case4gs	4	1.83e-03	2.13e-02	9.27e-02	4.51e-01	3.23e-03	2.94e-02	3.57e-01	3.52e-01
case5	5	4.85e-03	7.29e-02	5.13e-02	6.95e-01	7.33e-03	6.25e-02	3.26e-01	3.32e-01
case6ww	6	7.56e-04	2.34e-02	6.25e-01	1.47e-01	6.81e-04	1.73e-02	2.41e-01	2.01e-01
case9	9	2.76e-03	2.62e-02	4.77e-01	2.40e-01	2.40e-03	2.18e-02	1.28e-01	1.23e-01
case14	14	7.69e-04	2.28e-02	4.77e-01	4.18e-01	9.55e-04	1.68e-02	5.34e-02	7.73e-02
case24_ieee_rts	24	2.73e-03	1.24e-01	4.93e-01	1.48e+00	3.91e-03	5.80e-02	2.86e-01	3.82e-01
GBreducednetwork	29	3.03e-01	1.26e+03	8.28e+00	9.69e+01	2.26e-01	3.48e+00	1.94e+00	4.69e+02
case30	30	1.35e-03	1.41e-02	1.23e-01	9.04e-02	9.44e-04	1.18e-02	5.87e-02	4.35e-02
case_ieee30	30	8.33e-04	3.01e-02	1.57e-01	3.07e-01	8.13e-04	2.55e-02	5.22e-02	8.64e-02
case33bw	33	3.96e-03	2.85e-03	3.04e-02	1.02e-01	3.66e-03	4.04e-03	2.08e-02	4.78e-02
case39	39	8.76e-04	2.69e-01	8.25e-01	2.25e+00	1.39e-03	2.78e-01	3.44e-01	6.32e-01
case57	57	2.52e-02	3.63e-02	3.22e-01	7.02e-01	4.13e-02	2.19e-02	2.10e-01	3.00e-01
case89pegase	89	1.68e-01	5.72e-01	2.79e-01	5.24e+00	1.29e+00	6.13e-01	4.40e-01	1.68e+00
SimBench	94	6.68e-03	1.47e-01	1.34e+00	3.53e+00	7.49e-03	1.50e-01	7.07e-01	1.39e+00
case118	118	3.21e-03	1.13e-01	1.07e+00	1.28e+00	1.99e-03	6.69e-02	4.22e-01	2.97e-01
case145	145	2.05e-01	3.05e+01	1.79e+01	1.90e+02	1.10e-01	1.24e+01	8.54e-01	4.14e+02
iceland	189	6.83e+02	5.28e-01	1.16e-01	2.10e+00	1.33e+03	4.33e-01	5.59e-01	6.03e-01
case_illinois200	200	1.06e-03	1.20e-01	4.60e-01	8.64e-01	7.74e-04	1.20e-01	1.30e-01	1.64e-01
case300	300	9.92e-02	2.12e+00	3.78e-01	9.57e+00	2.50e-01	1.10e+00	3.60e-01	2.15e+00
LVN_heo1	722	NAN	3.21e-01	4.24e-02	1.18e-01	NAN	3.19e-01	4.03e-01	3.79e-01
case1354pegase	1354	1.75e+03	3.44e+00	4.17e-01	4.45e+00	2.68e+03	1.98e+00	3.39e-01	1.46e+00
case1888rte	1888	NAN	4.45e+00	1.48e+00	4.32e+00	NAN	2.20e+00	9.81e-01	1.35e+00
GBnetwork	2224	NAN	6.85e+00	3.04e-01	8.27e+01	NAN	2.74e+00	9.13e-01	1.56e+01
case2848rte	2848	NAN	1.63e+00	3.34e-01	2.50e+00	NAN	9.14e-01	4.74e-01	1.07e+00
case2869pegase	2869	1.96e+03	1.62e+01	5.44e-01	3.02e+02	3.63e+03	6.53e+00	3.41e-01	7.15e+01
case3120sp	3120	2.44e+03	4.61e+00	2.10e-01	7.08e+00	3.39e+03	2.01e+00	7.53e-01	1.72e+00
ENTSO_E_RealGridTest	6051	—	1.18e+00	1.42e-01	2.71e+00	—	6.77e-01	7.05e-01	8.87e-01
case6470rte	6470	—	3.60e+00	2.24e-01	5.71e+00	—	1.63e+00	9.08e-01	1.51e+00
case6495rte	6495	—	2.04e+00	7.34e-01	3.68e+00	—	1.06e+00	1.03e+00	1.33e+00
case6515rte	6515	—	2.51e+00	4.79e-01	4.48e+00	—	1.18e+00	9.83e-01	1.29e+00
case9241pegase	9241	—	3.62e+01	3.95e-01	2.90e+03	—	1.75e+01	1.45e+00	7.87e+02

Table 11: 95th-percentile residual, pu.

Where the last two families come from. The four drivers all compute the median and the tolerance fraction, but only the PIGNN driver printed them; `format_residual_distribution_compact` in the GridFM-derived drivers reported mean, p95 and RMSE only. The two families below were therefore recovered by re-running each saved checkpoint through its own driver with `--EPOCHS 0 --init_checkpoint`, which skips training and repeats the test pass on the same split. The rescore reproduces the published figures exactly — graphkit on ENTSO_E gives $|V| = 2.4238e-02$ against 2.42e-02 in the table above, on case6470rte 3.9982e-02 against 4.00e-02, and lumina on case118 2.5184e-02 against 2.52e-02 — which is the check that the split and the weights line up.

case9241pegase, and four cells that were blank. The largest grid had no row at all: every model hit the 24h wall before its test pass. Its checkpoints survived, so the row above is the `_v2t` checkpoint — the v2 campaign’s own resubmission, the same source as the other 6000-bus rows — rescored. The same pass filled `gridsfm` on ENTSO_E, case6470rte, case6495rte and case6515rte. PIGNN keeps a dash on all five: it produced no checkpoint on any of them.

For context, `acpf_extended-training_results` reports these two models on case9241pegase after a longer DDP run — graphkit 3.90e-02 at 300 epochs, gridsfm 3.92e-02 at 125 — against 5.32e-02 and 4.00e-02 here. Those are a different epoch budget and are deliberately kept out of this table, whose every other row is a 40-epoch v2 run.

Grid	Buses	median $ \Delta P $ [pu]				median $ \Delta Q $ [pu]			
		pigmn	graphkit	gridsfm	lumina	pigmn	graphkit	gridsfm	lumina
case4gs	4	6.27e-04	5.88e-03	3.36e-02	1.97e-01	1.68e-03	1.58e-02	1.17e-01	1.01e-01
case5	5	2.33e-03	2.03e-02	1.49e-02	2.20e-01	3.64e-03	1.88e-02	1.03e-01	1.02e-01
case6ww	6	2.53e-04	6.13e-03	1.64e-01	4.10e-02	2.09e-04	4.74e-03	8.51e-02	6.28e-02
case9	9	8.65e-04	6.96e-03	3.31e-02	7.92e-02	7.63e-04	6.37e-03	3.54e-02	3.40e-02
case14	14	1.26e-04	5.37e-03	1.64e-02	5.28e-02	2.00e-04	5.08e-03	1.14e-02	2.52e-02
case24_ieee_rts	24	8.33e-04	2.81e-02	8.69e-02	3.32e-01	1.07e-03	1.72e-02	8.59e-02	1.12e-01
GBreducednetwork	29	2.00e-02	1.58e+00	6.31e-01	4.60e+00	6.78e-02	8.48e-01	2.84e-01	1.26e+00
case30	30	2.46e-04	3.71e-03	1.11e-02	1.35e-02	1.88e-04	3.43e-03	8.35e-03	7.26e-03
case_ieee30	30	9.35e-05	6.58e-03	1.07e-02	6.57e-02	1.79e-04	6.29e-03	6.68e-03	1.65e-02
case33bw	33	3.47e-04	6.58e-04	8.40e-03	2.08e-03	3.97e-04	6.67e-04	6.68e-03	1.28e-03
case39	39	2.39e-04	6.71e-02	1.38e-01	5.37e-01	4.26e-04	7.23e-02	9.19e-02	1.54e-01
case57	57	1.77e-03	8.66e-03	2.59e-02	1.37e-01	2.33e-03	6.22e-03	2.19e-02	3.50e-02
case89pegase	89	1.47e-02	1.33e-01	6.02e-02	9.98e-01	1.90e-02	9.97e-02	7.26e-02	3.68e-01
SimBench	94	8.86e-04	2.96e-02	2.76e-01	3.76e-01	8.54e-04	2.84e-02	7.15e-02	1.43e-01
case118	118	4.16e-04	2.55e-02	6.70e-02	1.59e-01	5.29e-04	1.61e-02	1.02e-01	6.03e-02
case145	145	1.57e-02	2.33e+00	3.76e-01	9.93e+00	9.60e-03	9.78e-01	1.81e-01	3.54e+00
iceland	189	1.09e+00	6.64e-02	1.11e-02	3.55e-01	2.19e+00	6.58e-02	4.26e-02	9.48e-02
case_illinois200	200	2.47e-04	2.37e-02	1.41e-02	1.48e-01	1.83e-04	2.38e-02	3.24e-02	4.17e-02
case300	300	8.43e-03	2.94e-01	4.72e-02	—	1.48e-02	1.31e-01	5.86e-02	—
LVN_heo1	722	—	4.66e-02	1.92e-03	—	—	4.35e-02	1.18e-02	—
case1354pegase	1354	6.88e+01	5.69e-01	4.17e-02	—	5.91e+01	2.40e-01	8.12e-02	—
case1888rte	1888	—	5.19e-01	—	—	—	2.86e-01	—	—
GBnetwork	2224	—	4.58e-01	4.32e-02	1.74e+00	—	1.91e-01	1.56e-01	4.40e-01
case2848rte	2848	—	1.88e-01	3.27e-02	2.04e-01	—	1.20e-01	8.51e-02	8.62e-02
case2869pegase	2869	6.51e+01	2.02e+00	4.43e-02	1.48e+01	8.03e+01	5.39e-01	8.17e-02	2.53e+00
case3120sp	3120	1.22e+02	4.27e-01	2.17e-02	5.85e-02	1.32e+02	2.37e-01	4.04e-02	1.83e-02
ENTSO_E_RealGridTest	6051	—	1.08e-01	1.35e-02	9.55e-02	—	7.34e-02	5.91e-02	4.23e-02
case6470rte	6470	—	2.22e-01	2.65e-02	1.78e-01	—	1.10e-01	6.78e-02	6.01e-02
case6495rte	6495	—	1.28e-01	6.52e-02	1.71e-01	—	8.56e-02	4.92e-02	5.69e-02
case6515rte	6515	—	1.82e-01	4.96e-02	1.64e-01	—	9.63e-02	4.94e-02	5.44e-02
case9241pegase	9241	—	3.49e+00	3.74e-02	2.32e+01	—	1.22e+00	1.56e-01	4.74e+00

Table 12: Median power-balance residual, pu on a 100 MVA base, over the same scored buses as the mean. The mean and mean-scenario-maximum columns are set by a small number of pathological buses; the median describes the bulk, and the two can disagree by orders of magnitude on the ill-conditioned grids.

Grid	Buses	frac $ \Delta P \leq 10^{-6}$				frac $ \Delta Q \leq 10^{-6}$			
		pignn	graphkit	gridsfm	lumina	pignn	graphkit	gridsfm	lumina
case4gs	4	0.09%	0.02%	0.00%	0.00%	0.02%	0.00%	0.00%	0.00%
case5	5	0.01%	0.00%	0.00%	0.00%	0.01%	0.00%	0.00%	0.00%
case6ww	6	0.22%	0.01%	0.00%	0.00%	0.24%	0.01%	0.00%	0.00%
case9	9	0.04%	0.01%	0.00%	0.00%	0.07%	0.00%	0.00%	0.00%
case14	14	0.44%	0.01%	0.00%	0.00%	0.26%	0.01%	0.00%	0.00%
case24_ieee_rts	24	0.06%	0.00%	0.00%	0.00%	0.06%	0.00%	0.00%	0.00%
GBreducednetwork	29	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
case30	30	0.23%	0.01%	3.47%	0.00%	0.31%	0.01%	0.01%	0.00%
case_ieee30	30	0.52%	0.01%	0.00%	0.00%	0.21%	0.01%	0.01%	0.00%
case33bw	33	0.11%	0.11%	0.01%	0.01%	0.12%	0.09%	0.01%	0.00%
case39	39	0.21%	0.00%	0.00%	0.00%	0.12%	0.00%	0.00%	0.00%
case57	57	0.06%	0.01%	0.00%	0.00%	0.04%	0.01%	0.00%	0.00%
case89pegase	89	0.03%	0.00%	0.00%	0.00%	0.01%	0.00%	0.00%	0.00%
SimBench	94	0.06%	0.00%	0.00%	0.00%	0.07%	0.00%	0.00%	0.00%
case118	118	0.15%	0.00%	0.00%	0.00%	0.06%	0.00%	0.00%	0.00%
case145	145	0.00%	0.00%	0.49%	0.00%	0.01%	0.00%	0.00%	0.00%
iceland	189	0.03%	0.01%	0.03%	0.00%	0.03%	0.01%	0.00%	0.05%
case_illinois200	200	0.23%	0.00%	0.02%	0.00%	0.31%	0.00%	0.00%	0.00%
case300	300	0.01%	0.00%	0.01%	—	0.00%	0.00%	0.00%	—
LVN_heo1	722	—	0.00%	1.29%	—	—	0.00%	0.03%	—
case1354pegase	1354	0.00%	0.00%	0.00%	—	0.00%	0.00%	0.00%	—
case1888rte	1888	—	0.00%	—	—	—	0.00%	—	—
GBnetwork	2224	—	0.00%	0.03%	0.09%	—	0.00%	0.00%	0.00%
case2848rte	2848	—	0.00%	0.25%	0.00%	—	0.00%	0.00%	0.00%
case2869pegase	2869	0.00%	0.00%	0.11%	0.00%	0.00%	0.00%	0.00%	0.00%
case3120sp	3120	0.00%	0.00%	0.01%	0.01%	0.00%	0.00%	0.00%	0.00%
ENTSO_E_RealGridTest	6051	—	0.00%	0.01%	0.01%	—	0.00%	0.00%	0.01%
case6470rte	6470	—	0.00%	0.01%	0.00%	—	0.00%	0.00%	0.02%
case6495rte	6495	—	0.00%	0.11%	0.00%	—	0.00%	0.00%	0.00%
case6515rte	6515	—	0.00%	0.43%	0.01%	—	0.00%	0.01%	0.00%
case9241pegase	9241	—	0.00%	0.98%	0.00%	—	0.00%	0.00%	0.00%

Table 13: Fraction of scored buses already inside the Newton tolerance of 10^{-6} pu; higher is better, so **bold** marks the maximum. This is the strictest reading of the corpus: outside the smallest grids almost no bus is converged by this standard, and on every grid at or above 300 buses the figure is zero for all four models.

5 R_σ^2 : how far is the prediction from the solver?

All R^2 in this section are R_σ^2 of Section 6: $1 - (\text{RMSE}/\sigma)^2$ against a per-bus constant predictor. It is not a tracking measure — it also charges the model for bias and amplitude error — and the tracking coefficient R_{bus}^2 is reported separately there.

RMSE says how far a prediction sits from the reference. It does not say whether the model responds to its input. Bus voltage magnitude occupies a narrow band, so a model that ignores the scenario and emits the corpus-mean profile still scores an RMSE equal to the spread of the data. On this corpus that spread, σ , is between 1.5×10^{-3} and 4.7×10^{-2} pu depending on the grid — squarely inside the range several models report.

The coefficient of determination separates the two. Pooling over all buses and all test scenarios,

$$R_\sigma^2 = 1 - \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}} = 1 - \left(\frac{\text{RMSE}}{\sigma} \right)^2,$$

which is exact, not an approximation: $\text{SS}_{\text{res}} = N \text{RMSE}^2$ and $\text{SS}_{\text{tot}} = N \sigma^2$ by definition. $R_\sigma^2 = 1$ is a perfect fit, $R_\sigma^2 = 0$ is exactly as good as the per-bus constant predictor, and $R_\sigma^2 < 0$ is worse than it.

σ was measured directly from the reference solutions of each grid (300 scenarios, all buses); angles are taken about the circular mean so a bus near $\pm\pi$ does not acquire a spurious spread. No model is involved in σ .

Grid	Buses	σ	R_σ^2 on $ V $			
			pignn	graphkit	gridsfm	lumina
case4gs	4	1.966e-02	1.000	1.000	0.743	0.161
case5	5	2.482e-02	1.000	1.000	0.996	0.018
case6ww	6	2.580e-02	1.000	1.000	0.946	0.620
case9	9	2.309e-02	1.000	0.999	0.963	0.235
case14	14	2.313e-02	1.000	0.998	0.981	0.045
case24_ieee_rts	24	2.486e-02	1.000	0.998	0.916	-0.003
GBreducednetwork	29	1.663e-03	0.999	$\ll 0$	$\ll 0$	$\ll 0$
case30	30	2.204e-02	1.000	0.997	0.995	0.004
case_ieee30	30	2.354e-02	1.000	0.992	0.983	0.285
case33bw	33	1.520e-02	0.936	0.991	0.886	-0.371
case39	39	2.725e-02	1.000	0.928	0.893	-0.109
case57	57	3.553e-02	0.708	0.983	0.693	-0.027
case89pegase	89	2.399e-02	0.987	0.797	0.305	-0.523
SimBench	94	4.735e-02	0.999	0.988	0.942	-0.310
case118	118	2.469e-02	1.000	0.997	0.376	-0.042
case145	145	2.726e-02	0.884	$\ll 0$	-2.100	-5.040
iceland	189	3.128e-02	$\ll 0$	0.545	-1.190	-0.627
case_illinois200	200	2.074e-02	1.000	0.848	0.932	-0.317
case300	300	3.244e-02	0.343	0.466	0.420	-0.701
LVN_heo1	722	1.766e-02	—	-0.308	-2.060	-2.060
case1354pegase	1354	2.600e-02	$\ll 0$	0.354	0.046	-0.323
case1888rte	1888	2.075e-02	—	-0.475	-0.953	-1.514
GBnetwork	2224	2.365e-02	—	-1.241	-0.820	-2.525
case2848rte	2848	1.767e-02	—	-1.233	0.107	-4.125
case2869pegase	2869	2.767e-02	$\ll 0$	-0.346	0.605	-1.228
case3120sp	3120	2.106e-02	$\ll 0$	-0.844	-1.097	-1.560
ENTSO_E_RealGridTest	6051	1.888e-02	—	-0.643	—	-2.359
case6470rte	6470	3.142e-02	—	-0.621	—	-0.882
case6495rte	6495	3.273e-02	—	-0.398	—	-0.857
case6515rte	6515	3.106e-02	—	-0.505	—	-1.025
case9241pegase	9241	2.399e-02	—	-3.916	-1.774	-5.289

Table 14: R_σ^2 on voltage magnitude, pooled over buses and test scenarios. σ is the RMS spread of the reference $|V|$ about its mean, in pu — equivalently the RMSE a constant predictor achieves. Best per grid in bold; $\ll 0$ marks a model more than ten times worse than the mean predictor. NAN runs and missing runs are shown as —.

Grid	Buses	σ	R_σ^2 on θ			
			pignn	graphkit	gridsfm	lumina
case4gs	4	0.615	1.000	0.998	0.871	0.177
case5	5	0.679	1.000	0.992	0.998	0.401
case6ww	6	1.43	1.000	0.998	-3.794	0.683
case9	9	1.84	0.999	0.996	$\ll 0$	0.435
case14	14	3.46	1.000	0.999	0.499	0.886
case24_ieee_rts	24	3.34	1.000	0.994	0.765	0.177
GBreducednetwork	29	12.4	1.000	0.319	-6.004	0.057
case30	30	0.865	0.997	0.987	-3.529	-0.079
case_ieee30	30	4	1.000	0.999	0.068	0.708
case33bw	33	0.13	-3.166	0.383	$\ll 0$	-4.127
case39	39	5.83	1.000	0.992	-9.402	0.024
case57	57	4.1	0.999	0.997	0.443	0.190
case89pegase	89	2.25	0.984	0.908	0.778	-7.570
SimBench	94	6.54	1.000	0.998	0.447	-0.268
case118	118	4.24	1.000	0.988	$\ll 0$	0.255
case145	145	86.7	0.997	0.955	-0.357	0.378
iceland	189	11.2	$\ll 0$	0.950	-4.199	0.305
case_illinois200	200	2.45	0.999	0.949	$\ll 0$	-0.067
case300	300	19.3	0.981	0.939	-1.056	0.033
LVN_heo1	722	1.06	—	-0.618	0.502	-1.973
case1354pegase	1354	4.37	$\ll 0$	-0.170	-9.694	-4.765
case1888rte	1888	8.49	—	0.284	$\ll 0$	-1.879
GBnetwork	2224	7	—	0.502	0.906	-5.042
case2848rte	2848	3.84	—	0.341	$\ll 0$	-1.224
case2869pegase	2869	14.6	$\ll 0$	0.558	$\ll 0$	-2.881
case3120sp	3120	8.48	$\ll 0$	0.104	-1.102	-2.737
ENTSO_E_RealGridTest	6051	8.24	—	0.747	—	-0.781
case6470rte	6470	15.2	—	0.360	—	-0.302
case6495rte	6495	6.45	—	0.724	—	-2.288
case6515rte	6515	10.2	—	0.386	—	-1.028

Table 15: R_σ^2 on voltage angle, with σ in degrees about the circular mean. case33bw is a radial distribution feeder whose angles barely move ($\sigma = 0.13^\circ$), so every model scores poorly there on a quantity of almost no physical consequence; case145 sits at the other extreme ($\sigma = 87^\circ$), where a large absolute error is still a modest relative one.

What the table says. PIGNN is the only model that reaches $R_\sigma^2 > 0.99$ on $|V|$, and it does so on most grids below 200 buses. graphkit follows, clearing 0.9 on the small grids and decaying steadily with size. gridsfm and lumina sit near or below zero almost everywhere: on the majority of grids they are no better than, or worse than, predicting the mean voltage profile. That is the failure the RMSE tables cannot show, and it is the single most important qualification on the headline ranking. **But R_σ^2 cannot separate a model that fails to track from one that tracks and is miscalibrated,** and Section 7 shows the second is mostly what is happening: on R_{bus}^2 most of these cells are positive.

Above roughly 700 buses *no* model clears $R_\sigma^2 = 0.7$ on $|V|$. On R_{bus}^2 (Table 17) several clear 0.5 at that size, so the ordering in the RMSE tables there is a ranking of calibration as much as of skill.

Slope is now measured. R_σ^2 follows from RMSE, so it was available for every cell above; the regression slope needs the predictions themselves and therefore a pass over each saved checkpoint. Those passes have since run for all four models on all 31 grids, and the results are in Section 7. A slope near zero with R_{bus}^2 near zero is collapse; a slope near one with low R_{bus}^2 is an unbiased but noisy model; a slope well below one with a high R_{bus}^2 — the common case here — is a model that tracks and under-states the spread. R_σ^2 settles none of the three.

6 Notation: three different R^2 , and the calibration

Three distinct coefficients appear in the literature and in this project’s logs, and earlier revisions of this report used the word “ R^2 ” for more than one of them. Every table below names which one it is. Write v for the reference $|V|$ at a bus in a scenario, $\hat{v}$ for the prediction, $\bar{v}_i$ for bus i ’s mean over scenarios, and fit $\hat{v} = a v + b + \varepsilon$ by least squares.

Symbol	Definition	Null model it scores 0 against	Calibration
R_{pool}^2	Pearson R^2 of $\hat{v}$ on v , pooled over all buses and scenarios	one number for every bus and scenario	changes
R_{bus}^2	the same after subtracting $\bar{v}_i$ from $\hat{v}$ and v alike	each bus at its own mean	invariant
R_{σ}^2	$1 - \sum(\hat{v} - v)^2 / \sum(v - \bar{v}_i)^2$	each bus at its own mean	changes

R_{pool}^2 is the diagnostic GridSFM’s section 6.2 defines, and it is the right one for a multi-grid foundation model, which must reproduce a static voltage profile it has never seen. Every campaign here trains one model per grid on that grid, where that profile is memorisable from the training split; a predictor that emits each bus’s training mean and ignores the scenario entirely scores $R_{\text{pool}}^2 = 0.798$ on GBnetwork. So R_{bus}^2 is the tracking question in this setting, and R_{pool}^2 is kept only for comparability with the published numbers.

R_{σ}^2 is what earlier revisions of this report printed. It is not a tracking measure: it charges the model for bias and for amplitude error as well. Writing $\sigma^2 = \frac{1}{N} \sum (v - \bar{v}_i)^2$ and σ_{ε}^2 for the variance of ε , the mean square error splits exactly into three terms,

$$\text{RMSE}^2 = \underbrace{b^2}_{\text{bias}} + \underbrace{(a-1)^2 \sigma^2}_{\text{amplitude}} + \underbrace{\sigma_{\varepsilon}^2}_{\text{noise}}, \quad \text{so} \quad R_{\sigma}^2 = 1 - \frac{b^2}{\sigma^2} - (a-1)^2 - \frac{\sigma_{\varepsilon}^2}{\sigma^2}.$$

R_{bus}^2 normalises the noise term by the variance of the *prediction* rather than of the reference, so it is not a simple difference away from R_{σ}^2 ; the two answer different questions and should not be converted into one another.

Three relations are used below, and each was checked against the measurements rather than assumed. (i) Calibration removes exactly the b^2/σ^2 term, so $R_{\sigma, \text{cal}}^2 = 1 - (a-1)^2 - \sigma_{\varepsilon}^2/\sigma^2$; over the 26 stabilization runs this held to 1.4×10^{-2} . (ii) R_{bus}^2 and the slope are unchanged by calibration, since a per-bus offset cannot alter a regression on per-bus deviations; measured deviation $< 10^{-6}$ on all 26. (iii) $R_{\sigma, \text{cal}}^2 \leq R_{\text{bus}}^2$, with the gap being the amplitude error; observed on all 26, from 0.0006 to 1.22.

Post-hoc per-bus calibration. The b^2/σ^2 term is removable without touching the model. For each bus, take the mean prediction over the *training* split, align it to the training reference mean, and subtract that fixed offset from every unseen test scenario:

$$\hat{v}_{i,s}^{\text{cal}} = \hat{v}_{i,s} - (\bar{\hat{v}}_i^{\text{train}} - \bar{v}_i^{\text{train}}),$$

and the circular equivalent for θ . No test data enters the fit, so this is a correction a deployed model could carry. Because it is fitted on one split and applied to another it does not transfer perfectly: across the 26 stabilization runs the realised improvement in R_{σ}^2 differs from the fitted b^2/σ^2 by a median of 1.2% and at most 8.4%. Tables that report it use the form *raw* $\rightarrow$ *calibrated*. R_{bus}^2 and the slope are mathematically unchanged by it — a per-bus offset cannot alter a regression on per-bus deviations — which is used below as a consistency check on the implementation.

7 Re-scored: what the R^2 tables above are and are not

The R^2 tables above are $1 - (\text{RMSE}/\sigma)^2$ with σ measured per bus, which is the right null for a per-grid surrogate but is *not* the diagnostic GridSFM’s section 6.2 defines. That one is a pooled regression of prediction against reference reported with a slope, and its R^2 is a Pearson R^2 : it asks whether the model orders scenarios correctly, and is blind to bias and to amplitude error. The discussion below Table 5 previously read the SSE quantity as if it answered the regression question. It does not, and where the two disagree they disagree by a lot.

All 31 grids have been re-scored from the saved checkpoints with both, plus per-edge angle diagnostics (alex jobs 4125850/4126397; 112 model-grid cells, 3000 test scenarios each). The rescore reproduces the earlier figures to every printed digit where they overlap, and its independently measured σ agrees with the hand-measured σ of Table 5 to a median of 1.3%, which confirms the per-bus convention used throughout this report.

Reading of R^2 on $ V $ (median over grids)	pignn	graphkit	gridsfm	lumina
R^2_{pool} (as GridSFM publishes it)	1.000	0.828	0.720	0.182
R^2_{bus} (per-bus deviations)	1.000	0.636	0.651	0.082
R^2_{σ} (what the tables above print)	1.000	0.493	0.373	-0.628

Table 16: The same checkpoints under three readings. Pooling over buses — as GridSFM does, correctly, for a model that must predict grids it has never seen — credits a per-grid surrogate for reproducing a static voltage profile it can memorise from its own training split. On GBnetwork a predictor that emits each bus’s training mean and ignores the scenario entirely scores 0.798 pooled. The middle row removes that profile; the bottom row is what the tables above print.

Does the model track the scenario? (R^2_{bus})

Grid	Buses	R^2_{bus} on $ V $ — does the model track the scenario?			
		pignn	graphkit	gridsfm	lumina
case4gs	4	1.000	1.000	0.874	0.120
case5	5	1.000	1.000	0.997	0.003
case6ww	6	1.000	1.000	0.981	0.643
case9	9	1.000	0.999	0.969	0.298
case14	14	1.000	0.998	0.992	0.082
case24_ieee_rts	24	1.000	0.998	0.934	0.127
GBreducednetwork	29	1.000	0.601	0.042	0.075
case30	30	1.000	0.999	0.996	0.119
case_ieee30	30	1.000	0.992	0.991	0.308
case33bw	33	0.985	0.992	0.912	0.002
case39	39	1.000	0.938	0.946	0.166
case57	57	0.828	0.986	0.731	0.305
case89pegase	89	0.991	0.811	0.318	0.039
SimBench	94	0.999	0.990	0.995	0.240
case118	118	1.000	0.998	0.674	0.002
case145	145	0.936	0.130	0.065	0.000
iceland	189	—	0.625	0.298	0.136
case_illinois200	200	1.000	0.858	0.936	0.115
case300	300	0.733	0.636	0.513	0.201
LVN_heo1	722	—	0.088	0.024	0.000
case1354pegase	1354	0.007	0.550	0.165	0.025
case1888rte	1888	—	0.395	0.589	0.003
GBnetwork	2224	—	0.232	0.752	0.010
case2848rte	2848	—	0.535	0.464	0.012
case2869pegase	2869	—	0.302	0.651	0.001
case3120sp	3120	—	0.338	0.047	0.004
ENTSO_E_RealGridTest	6051	—	0.435	0.587	0.121
case6470rte	6470	—	0.148	0.480	0.099
case6495rte	6495	—	0.293	0.471	0.052
case6515rte	6515	—	0.137	0.447	0.090
case9241pegase	9241	—	0.117	0.081	0.000
median		1.000	0.636	0.651	0.082

Grid	Buses	R^2_{bus} on $ V $ — does the model track the scenario?			
		pignn	graphkit	gridsfm	lumina

Table 17: Pearson R^2 of predicted against reference $|V|$ after each bus’s own mean is removed from both. Best per grid in bold; — marks a grid with no v2 checkpoint for that model. **This is a materially more favourable picture than the SSE tables above.** graphkit and gridsfm hold median 0.64 and 0.65, and on several large grids where Table 5 reports a strongly negative R^2 they still explain half the scenario variance — gridsfm on GBnetwork 0.752 against -0.820 , graphkit on case1888rte 0.395 against -0.475 . lumina is the exception, and the pooled column was flattering it: median 0.082, and at or below 0.01 on eleven grids, which is very nearly a fixed profile.

Grid	Buses	regression slope on $ V $, per-bus deviations (best is nearest 1)			
		pignn	graphkit	gridsfm	lumina
case4gs	4	1.000	1.001	0.655	0.097
case5	5	1.000	1.000	0.991	0.003
case6ww	6	1.000	0.999	0.950	0.566
case9	9	1.000	0.993	0.962	0.248
case14	14	1.000	0.999	0.974	0.068
case24_ieee_rts	24	1.000	0.989	0.887	0.059
GBreducednetwork	29	1.005	-0.228	-1.614	1.135
case30	30	1.000	1.002	0.988	0.000
case_ieee30	30	0.998	0.979	0.978	0.281
case33bw	33	0.876	0.985	0.919	0.000
case39	39	1.000	0.897	0.806	0.056
case57	57	0.744	0.964	0.709	0.423
case89pegase	89	0.935	0.680	0.283	0.022
SimBench	94	0.992	0.976	1.015	0.035
case118	118	1.000	0.991	0.382	0.001
case145	145	0.904	0.832	0.268	-0.003
iceland	189	—	0.629	0.294	0.099
case_illinois200	200	0.999	0.828	0.911	0.057
case300	300	0.684	0.561	0.413	0.164
LVN_heo1	722	—	0.081	0.019	0.000
case1354pegase	1354	0.251	0.483	0.133	0.017
case1888rte	1888	—	0.423	0.536	0.004
GBnetwork	2224	—	0.150	0.689	0.007
case2848rte	2848	—	0.558	0.389	0.014
case2869pegase	2869	—	0.351	0.536	0.000
case3120sp	3120	—	0.321	0.050	0.015
ENTSO_E_RealGridTest	6051	—	0.265	0.584	0.114
case6470rte	6470	—	0.122	0.494	0.090
case6495rte	6495	—	0.177	0.408	0.019
case6515rte	6515	—	0.129	0.425	0.040
case9241pegase	9241	—	0.364	0.069	-0.000
median		1.000	0.680	0.536	0.035

Grid	Buses	regression slope on $ V $, per-bus deviations (best is nearest 1)			
		pignn	graphkit	gridsfm	lumina

Table 18: Regression slope on the same per-bus deviations. Bold is closest to 1, not largest. Slope is the amplitude question R^2 cannot answer: a model can order every scenario correctly and still emit a fraction of the true spread. Read against Table 17: a low slope with a high R^2 is under-confidence, correctable by rescaling; a low slope with a low R^2 is collapse.

Post-hoc calibration, corpus-wide

The per-bus offset of Section 6 was fitted on each model’s own training split and applied unchanged to the test split, for all 112 model-grid cells (alex job 4129152). Nothing is retrained. The invariance check passes: R^2_{bus} and the slope move by less than 10^{-6} on 111 of 112 cells.

Median over grids	pignn	graphkit	gridsfm	lumina
$ V $ RMSE, raw	0.0001	0.0202	0.0199	0.0329
$ V $ RMSE, calibrated	0.0001	0.0106	0.0133	0.0232
R^2_{σ} , raw	1.000	0.493	0.373	-0.628
R^2_{σ} , calibrated	1.000	0.626	0.587	0.031
R^2_{bus} (invariant)	1.000	0.636	0.651	0.082
slope (invariant)	1.000	0.680	0.536	0.035
R^2_{pool}	1.000	0.828	0.720	0.182
per-edge $\Delta\theta_{ij}$, raw	0.010	0.524	1.398	2.256
per-edge $\Delta\theta_{ij}$, calibrated	0.009	0.474	0.475	1.107
$R^2_{\sigma} > 0$, raw	18/19	18/31	19/31	6/31
$R^2_{\sigma} > 0$, calibrated	18/19	28/31	29/31	23/31

Table 19: What the calibration buys, corpus-wide. Median $|V|$ RMSE falls by $1.9\times$ for graphkit, $1.5\times$ for gridsfm and $1.4\times$ for lumina; R^2_{σ} rises to just under R^2_{bus} , as it must, the remaining gap being the amplitude error that the slope column reports. The count of grids on which a model beats the per-bus constant predictor goes from 18 to 28 for graphkit, 19 to 29 for gridsfm and 6 to 23 for lumina. pignn’s medians are dominated by the small grids it has checkpoints for and are not comparable across the row. Per-edge $\Delta\theta_{ij}$ also improves, most sharply for gridsfm ($1.398^\circ \rightarrow 0.475^\circ$), because the calibration includes a per-bus angle offset.

Grid	Buses	$ V $ RMSE [pu], raw $\rightarrow$ calibrated			
		pignn	graphkit	gridsfm	lumina
case4gs	4	0.0000 $\rightarrow$ 0.0000	0.0003 $\rightarrow$ 0.0002	0.0100 $\rightarrow$ 0.0081	0.0179 $\rightarrow$ 0.0179
case5	5	0.0000 $\rightarrow$ 0.0000	0.0001 $\rightarrow$ 0.0001	0.0016 $\rightarrow$ 0.0012	0.0246 $\rightarrow$ 0.0246
case6ww	6	0.0000 $\rightarrow$ 0.0000	0.0004 $\rightarrow$ 0.0004	0.0060 $\rightarrow$ 0.0037	0.0159 $\rightarrow$ 0.0159
case9	9	0.0001 $\rightarrow$ 0.0001	0.0007 $\rightarrow$ 0.0007	0.0045 $\rightarrow$ 0.0042	0.0202 $\rightarrow$ 0.0201
case14	14	0.0001 $\rightarrow$ 0.0001	0.0010 $\rightarrow$ 0.0010	0.0032 $\rightarrow$ 0.0022	0.0226 $\rightarrow$ 0.0225
case24_ieee_rts	24	0.0001 $\rightarrow$ 0.0001	0.0012 $\rightarrow$ 0.0011	0.0071 $\rightarrow$ 0.0066	0.0249 $\rightarrow$ 0.0239
GBreducednetwork	29	0.0000 $\rightarrow$ 0.0000	0.4090 $\rightarrow$ 0.0020	0.0524 $\rightarrow$ 0.0134	0.0601 $\rightarrow$ 0.0065
case30	30	0.0001 $\rightarrow$ 0.0001	0.0011 $\rightarrow$ 0.0008	0.0016 $\rightarrow$ 0.0014	0.0219 $\rightarrow$ 0.0219
case_ieee30	30	0.0002 $\rightarrow$ 0.0001	0.0021 $\rightarrow$ 0.0021	0.0030 $\rightarrow$ 0.0022	0.0199 $\rightarrow$ 0.0197
case33bw	33	0.0039 $\rightarrow$ 0.0025	0.0015 $\rightarrow$ 0.0013	0.0051 $\rightarrow$ 0.0045	0.0179 $\rightarrow$ 0.0151
case39	39	0.0001 $\rightarrow$ 0.0000	0.0073 $\rightarrow$ 0.0067	0.0089 $\rightarrow$ 0.0073	0.0286 $\rightarrow$ 0.0252
case57	57	0.0189 $\rightarrow$ 0.0152	0.0045 $\rightarrow$ 0.0043	0.0199 $\rightarrow$ 0.0185	0.0364 $\rightarrow$ 0.0307
case89pegase	89	0.0028 $\rightarrow$ 0.0026	0.0108 $\rightarrow$ 0.0106	0.0200 $\rightarrow$ 0.0192	0.0296 $\rightarrow$ 0.0229
SimBench	94	0.0015 $\rightarrow$ 0.0012	0.0053 $\rightarrow$ 0.0051	0.0115 $\rightarrow$ 0.0038	0.0552 $\rightarrow$ 0.0490
case118	118	0.0000 $\rightarrow$ 0.0000	0.0013 $\rightarrow$ 0.0011	0.0195 $\rightarrow$ 0.0166	0.0251 $\rightarrow$ 0.0246
case145	145	0.0074 $\rightarrow$ 0.0069	0.1245 $\rightarrow$ 0.0584	0.0477 $\rightarrow$ 0.0339	0.0667 $\rightarrow$ 0.0281

Grid	Buses	$ V $ RMSE [pu], <i>raw</i> $\rightarrow$ <i>calibrated</i>			
		pignn	graphkit	gridsfm	lumina
iceland	189	—	0.0213 $\rightarrow$ 0.0183	0.0465 $\rightarrow$ 0.0250	0.0401 $\rightarrow$ 0.0279
case_illinois200	200	0.0002 $\rightarrow$ 0.0001	0.0081 $\rightarrow$ 0.0076	0.0054 $\rightarrow$ 0.0051	0.0236 $\rightarrow$ 0.0192
case300	300	0.0262 $\rightarrow$ 0.0171	0.0237 $\rightarrow$ 0.0201	0.0246 $\rightarrow$ 0.0234	0.0422 $\rightarrow$ 0.0295
LVN_heo1	722	—	0.0202 $\rightarrow$ 0.0168	0.0309 $\rightarrow$ 0.0174	0.0310 $\rightarrow$ 0.0176
case1354pegase	1354	0.1716 $\rightarrow$ 0.0792	0.0209 $\rightarrow$ 0.0176	0.0254 $\rightarrow$ 0.0239	0.0299 $\rightarrow$ 0.0257
case1888rte	1888	—	0.0253 $\rightarrow$ 0.0161	0.0291 $\rightarrow$ 0.0133	0.0329 $\rightarrow$ 0.0206
GBnetwork	2224	—	0.0353 $\rightarrow$ 0.0208	0.0319 $\rightarrow$ 0.0117	0.0444 $\rightarrow$ 0.0232
case2848rte	2848	—	0.0263 $\rightarrow$ 0.0121	0.0167 $\rightarrow$ 0.0131	0.0399 $\rightarrow$ 0.0177
case2869pegase	2869	—	0.0322 $\rightarrow$ 0.0230	0.0174 $\rightarrow$ 0.0166	0.0414 $\rightarrow$ 0.0274
case3120sp	3120	—	0.0286 $\rightarrow$ 0.0172	0.0305 $\rightarrow$ 0.0206	0.0337 $\rightarrow$ 0.0214
ENTSO_E_RealGridTest	6051	—	0.0241 $\rightarrow$ 0.0153	0.0269 $\rightarrow$ 0.0124	0.0346 $\rightarrow$ 0.0181
case6470rte	6470	—	0.0399 $\rightarrow$ 0.0299	0.0328 $\rightarrow$ 0.0233	0.0430 $\rightarrow$ 0.0307
case6495rte	6495	—	0.0387 $\rightarrow$ 0.0289	0.0355 $\rightarrow$ 0.0245	0.0447 $\rightarrow$ 0.0329
case6515rte	6515	—	0.0383 $\rightarrow$ 0.0300	0.0391 $\rightarrow$ 0.0240	0.0442 $\rightarrow$ 0.0312
case9241pegase	9241	—	0.0532 $\rightarrow$ 0.0291	0.0400 $\rightarrow$ 0.0235	0.0602 $\rightarrow$ 0.0245
median		0.0001 $\rightarrow$ 0.0001	0.0202 $\rightarrow$ 0.0106	0.0199 $\rightarrow$ 0.0133	0.0329 $\rightarrow$ 0.0232

Table 20: $|V|$ RMSE in pu, *raw* $\rightarrow$ *calibrated*. Every cell improves or is unchanged; no cell is made worse by the correction.

Grid	Buses	R^2_σ on $ V $, <i>raw</i> $\rightarrow$ <i>calibrated</i>			
		pignn	graphkit	gridsfm	lumina
case4gs	4	+1.00 $\rightarrow$ +1.00	+1.00 $\rightarrow$ +1.00	+0.72 $\rightarrow$ +0.82	+0.11 $\rightarrow$ +0.12
case5	5	+1.00 $\rightarrow$ +1.00	+1.00 $\rightarrow$ +1.00	+1.00 $\rightarrow$ +1.00	-0.00 $\rightarrow$ +0.00
case6ww	6	+1.00 $\rightarrow$ +1.00	+1.00 $\rightarrow$ +1.00	+0.95 $\rightarrow$ +0.98	+0.63 $\rightarrow$ +0.63
case9	9	+1.00 $\rightarrow$ +1.00	+1.00 $\rightarrow$ +1.00	+0.96 $\rightarrow$ +0.97	+0.28 $\rightarrow$ +0.29
case14	14	+1.00 $\rightarrow$ +1.00	+1.00 $\rightarrow$ +1.00	+0.98 $\rightarrow$ +0.99	+0.07 $\rightarrow$ +0.08
case24_ieee_rts	24	+1.00 $\rightarrow$ +1.00	+1.00 $\rightarrow$ +1.00	+0.92 $\rightarrow$ +0.93	+0.02 $\rightarrow$ +0.09
GBreducednetwork	29	+1.00 $\rightarrow$ +1.00	-62161.55 $\rightarrow$ -0.54	-1018.62 $\rightarrow$ -65.60	-1342.22 $\rightarrow$ -14.84
case30	30	+1.00 $\rightarrow$ +1.00	+1.00 $\rightarrow$ +1.00	+0.99 $\rightarrow$ +1.00	-0.00 $\rightarrow$ -0.00
case_ieee30	30	+1.00 $\rightarrow$ +1.00	+0.99 $\rightarrow$ +0.99	+0.98 $\rightarrow$ +0.99	+0.29 $\rightarrow$ +0.30
case33bw	33	+0.93 $\rightarrow$ +0.97	+0.99 $\rightarrow$ +0.99	+0.88 $\rightarrow$ +0.91	-0.40 $\rightarrow$ -0.00
case39	39	+1.00 $\rightarrow$ +1.00	+0.92 $\rightarrow$ +0.94	+0.89 $\rightarrow$ +0.92	-0.17 $\rightarrow$ +0.09
case57	57	+0.72 $\rightarrow$ +0.82	+0.98 $\rightarrow$ +0.99	+0.69 $\rightarrow$ +0.73	-0.04 $\rightarrow$ +0.26
case89pegase	89	+0.99 $\rightarrow$ +0.99	+0.78 $\rightarrow$ +0.79	+0.25 $\rightarrow$ +0.31	-0.63 $\rightarrow$ +0.03
SimBench	94	+1.00 $\rightarrow$ +1.00	+0.99 $\rightarrow$ +0.99	+0.95 $\rightarrow$ +0.99	-0.19 $\rightarrow$ +0.06
case118	118	+1.00 $\rightarrow$ +1.00	+1.00 $\rightarrow$ +1.00	+0.37 $\rightarrow$ +0.55	-0.04 $\rightarrow$ +0.00
case145	145	+0.93 $\rightarrow$ +0.94	-20.14 $\rightarrow$ -3.66	-2.10 $\rightarrow$ -0.57	-5.07 $\rightarrow$ -0.08
iceland	189	—	+0.49 $\rightarrow$ +0.62	-1.43 $\rightarrow$ +0.30	-0.80 $\rightarrow$ +0.12
case_illinois200	200	+1.00 $\rightarrow$ +1.00	+0.84 $\rightarrow$ +0.86	+0.93 $\rightarrow$ +0.93	-0.38 $\rightarrow$ +0.08
case300	300	+0.36 $\rightarrow$ +0.73	+0.48 $\rightarrow$ +0.63	+0.44 $\rightarrow$ +0.49	-0.65 $\rightarrow$ +0.19
LVN_heo1	722	—	-0.32 $\rightarrow$ +0.09	-2.08 $\rightarrow$ +0.02	-2.09 $\rightarrow$ -0.00
case1354pegase	1354	-42.52 $\rightarrow$ -8.26	+0.35 $\rightarrow$ +0.54	+0.05 $\rightarrow$ +0.16	-0.32 $\rightarrow$ +0.02
case1888rte	1888	—	-0.50 $\rightarrow$ +0.39	-0.99 $\rightarrow$ +0.58	-1.55 $\rightarrow$ +0.00
GBnetwork	2224	—	-1.29 $\rightarrow$ +0.20	-0.87 $\rightarrow$ +0.75	-2.62 $\rightarrow$ +0.01
case2848rte	2848	—	-1.20 $\rightarrow$ +0.53	+0.11 $\rightarrow$ +0.45	-4.06 $\rightarrow$ +0.01
case2869pegase	2869	—	-0.39 $\rightarrow$ +0.29	+0.60 $\rightarrow$ +0.63	-1.29 $\rightarrow$ -0.00
case3120sp	3120	—	-0.84 $\rightarrow$ +0.34	-1.08 $\rightarrow$ +0.05	-1.55 $\rightarrow$ -0.03
ENTSO_E_RealGridTest	6051	—	-0.56 $\rightarrow$ +0.37	-0.95 $\rightarrow$ +0.59	-2.22 $\rightarrow$ +0.12
case6470rte	6470	—	-0.53 $\rightarrow$ +0.14	-0.03 $\rightarrow$ +0.48	-0.77 $\rightarrow$ +0.10
case6495rte	6495	—	-0.35 $\rightarrow$ +0.25	-0.13 $\rightarrow$ +0.46	-0.80 $\rightarrow$ +0.03
case6515rte	6515	—	-0.41 $\rightarrow$ +0.14	-0.47 $\rightarrow$ +0.45	-0.88 $\rightarrow$ +0.06

Grid	Buses	R_σ^2 on $ V $, <i>raw</i> $\rightarrow$ <i>calibrated</i>			
		pigmn	graphkit	gridsfm	lumina
case9241pegase	9241	—	-3.72 $\rightarrow$ -0.41	-1.66 $\rightarrow$ +0.08	-5.04 $\rightarrow$ -0.00
median		+1.00 $\rightarrow$ +1.00	+0.49 $\rightarrow$ +0.63	+0.37 $\rightarrow$ +0.59	-0.63 $\rightarrow$ +0.03

Table 21: R_σ^2 on $|V|$, *raw* $\rightarrow$ *calibrated*. The left-hand value of each pair is what Table 5 prints. **Most of the negative entries in that table are removable without touching the model.** The large grids are where it matters most: gridsfm on GBnetwork $-0.85 \rightarrow +0.75$, on case2869pegase $-0.35 \rightarrow +0.62$; graphkit on case2848rte $-1.14 \rightarrow +0.53$. What does not move is lumina’s amplitude — its slope stays at a median 0.035 — so its calibrated R_σ^2 tops out near zero on most grids, which is the correct reading of a model that emits a fixed profile.

A caution the slope column earns. On case30, lumina reports $R_{\text{bus}}^2 = 0.119$, which looks like partial tracking. Its predicted per-bus spread is 2.3×10^{-7} pu against a reference spread of 2.2×10^{-2} , and its slope is 3.7×10^{-6} : the model is emitting a constant profile, and the 0.119 is correlation measured inside floating-point noise. This is the one cell where the calibration invariance check exceeds 10^{-6} (it differs in the sixth decimal), for the same reason. A R_{bus}^2 without its slope is not interpretable.

The angle tables are largely measuring gauge drift

Branch flows depend on $\theta_i - \theta_j$. A model whose bus angles carry a common per-scenario offset scores an arbitrarily bad per-bus θ RMSE while predicting every flow exactly, and GridSFM’s paper raises precisely this objection to gridfm-graphkit’s angle column. It applies here with force, and most of all to gridsfm itself.

Angle error RMSE, deg (median over grids)	pigmn	graphkit	gridsfm	lumina
per-bus θ	0.051	1.353	12.577	6.586
gauge component	0.045	0.492	9.735	3.556
per-edge $\Delta\theta_{ij}$	0.010	0.524	1.398	2.256
inflation factor	3.4 $\times$	2.4 $\times$	8.7 $\times$	2.4 $\times$

Table 22: “gauge component” is the circular mean of the bus-angle error within each scenario; “inflation factor” is the median per-grid ratio of per-bus to per-edge RMSE. **gridsfm’s angle error is inflated 8.7 $\times$** : a median 12.58° per bus against 1.40° on the quantity the flows see, 9.74° of the former being a pure offset. The angle ranking in Table 5 is therefore not a ranking of flow accuracy, and the θ RMSE column of the headline tables should not be compared across models with different gauge behaviour.

Grid	Buses	per-edge $\Delta\theta_{ij}$ RMSE [deg]			
		pigmn	graphkit	gridsfm	lumina
case4gs	4	0.002	0.024	0.195	0.585
case5	5	0.004	0.047	0.027	0.442
case6ww	6	0.004	0.051	1.972	0.554
case9	9	0.017	0.066	2.197	0.718
case14	14	0.004	0.070	1.398	0.766
case24_ieee_rts	24	0.009	0.133	0.623	1.499
GBreducednetwork	29	0.038	11.091	5.509	8.573
case30	30	0.010	0.052	0.508	0.377
case_ieee30	30	0.004	0.075	1.146	1.146
case33bw	33	0.052	0.024	0.154	0.078
case39	39	0.002	0.141	2.168	1.351
case57	57	0.082	0.129	1.356	1.620

Grid	Buses	per-edge $\Delta\theta_{ij}$ RMSE [deg]			
		pignn	graphkit	gridsfm	lumina
case89pegase	89	0.126	0.486	0.326	3.933
SimBench	94	0.020	0.107	4.732	2.256
case118	118	0.005	0.187	2.033	1.445
case145	145	2.856	12.294	23.478	37.947
iceland	189	—	0.817	1.670	3.310
case_illinois200	200	0.009	0.159	2.902	1.256
case300	300	0.339	2.164	1.642	4.870
LVN_heo1	722	—	0.524	0.176	1.017
case1354pegase	1354	76.400	1.456	0.638	2.737
case1888rte	1888	—	1.659	2.573	3.808
GBnetwork	2224	—	3.531	0.446	10.856
case2848rte	2848	—	1.153	1.818	1.935
case2869pegase	2869	—	4.460	1.379	8.107
case3120sp	3120	—	3.374	0.861	5.397
ENTSO_E_RealGridTest	6051	—	0.927	0.413	3.010
case6470rte	6470	—	2.003	0.409	5.130
case6495rte	6495	—	1.324	2.562	3.650
case6515rte	6515	—	1.578	1.615	4.123
case9241pegase	9241	—	13.655	1.831	27.515
median		0.010	0.524	1.398	2.256

Table 23: Per-edge $\Delta\theta_{ij}$ RMSE in degrees over in-service branches; lowest per grid in bold. On this quantity gridsfm takes about half the large grids, having lost nearly all of them on per-bus θ .

A calibration defect, not collapse. With $\hat{v} = av + b + \varepsilon$, $\text{RMSE}^2 = b^2 + (a-1)^2\sigma^2 + \sigma_\varepsilon^2$, so the gap between the regression R^2 and the SSE R^2 is exactly $(a-1)^2 + b^2/\sigma^2$: amplitude error plus bias, and nothing about tracking. On GBnetwork, where this was pursued in detail (`acpf_gbnetwork_case300_global-context-report`), the bias term alone exceeds σ for every model, and subtracting a per-bus offset fitted on the training split — no retraining, no architecture change — cuts $|V|$ RMSE by 2.0–2.7 $\times$, turns every R^2 positive and improves median $|\Delta Q|$ by 3–4 $\times$. That correction has now been applied corpus-wide as well (Section 7); the tables earlier in this report stand as measured, but what they measure is a miscalibration a post-hoc per-bus offset largely removes, not the absence of a learned mapping.

8 PIGNN diverges on four grids

On LVN_heo1, case1888rte, GBnetwork and case2848rte the PIGNN run produced a constant validation RMSE, and the cause is *not* the v1 defect. The zero-grad guard that killed 11 runs in v1 fired **zero times** here — `--armijo_mode geometric_safe` kept the graph edge alive as intended. What happens instead is divergence:

```
Epoch 0 | train loss 1.1116e+04  rmse 1.0297e+00  (dPinf 1.126e+05 pu)
Epoch 40 | train loss nan          rmse nan          (dPinf nan)
```

The initial loss is already 1.1×10^4 with a mean scenario-maximum mismatch of 1.1×10^5 pu, and the run reaches NaN. The reported validation RMSE is constant because the best checkpoint never moves past epoch 0. These cells are marked NAN and excluded from the aggregates.

This is a different failure from v1 and needs a different fix. The v1 problem was no gradient; here the gradient exists and is too large. `geometric_safe` always applies a step, including one that increases the residual, which plausibly contributes — it is the trade-off noted when the mode was chosen. Gradient clipping or a lower learning rate on these grids is the obvious next experiment; neither has been run.

9 Caveats

Coverage is incomplete. 22 of 31 grids have all four models: graphkit and lumina stand at 30 usable, gridsfm at 26, pigmn at 22. The gap is the five largest grids, whose runs are described above.

Twelve cells will carry a smaller budget. The resubmitted runs on the five largest grids use 11–30 epochs against 40 elsewhere, because 40 does not fit a 24 h limit at their measured cost and the 48 h alternative is preempted.

Single seed. Every number is one run. Differences below roughly a factor of two should not be read as real; PIGNN’s margins are far above that threshold, the graphkit–gridsfm gap on $|V|$ is not.